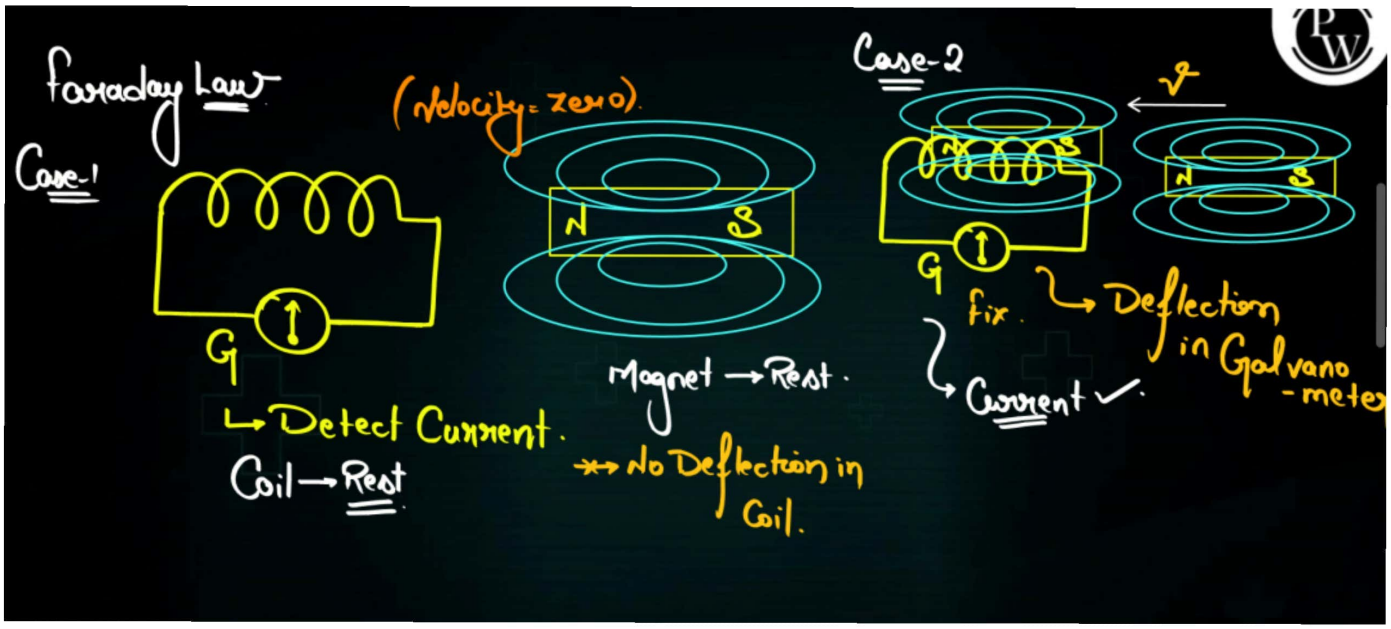
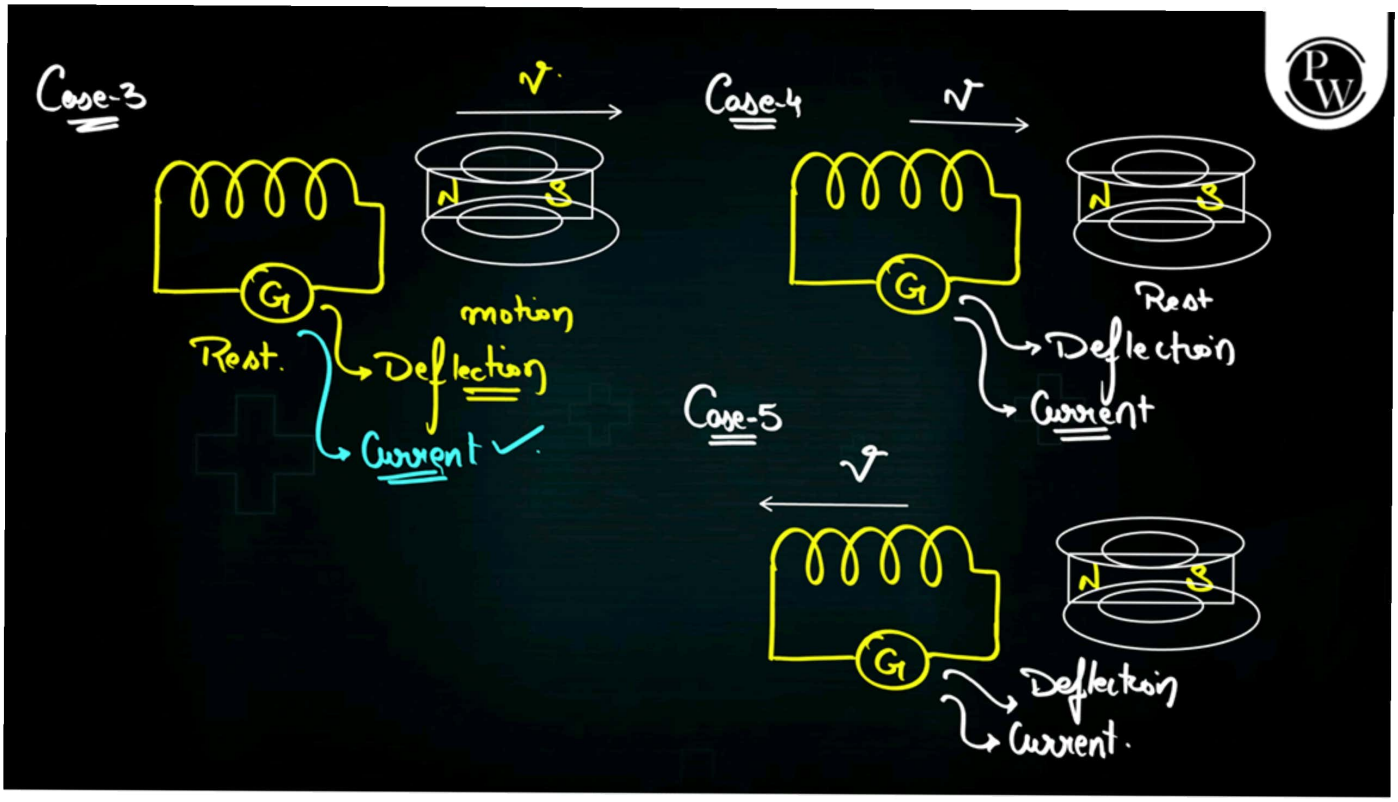
12th BOARD HACKER JUGAAD NOTES :-

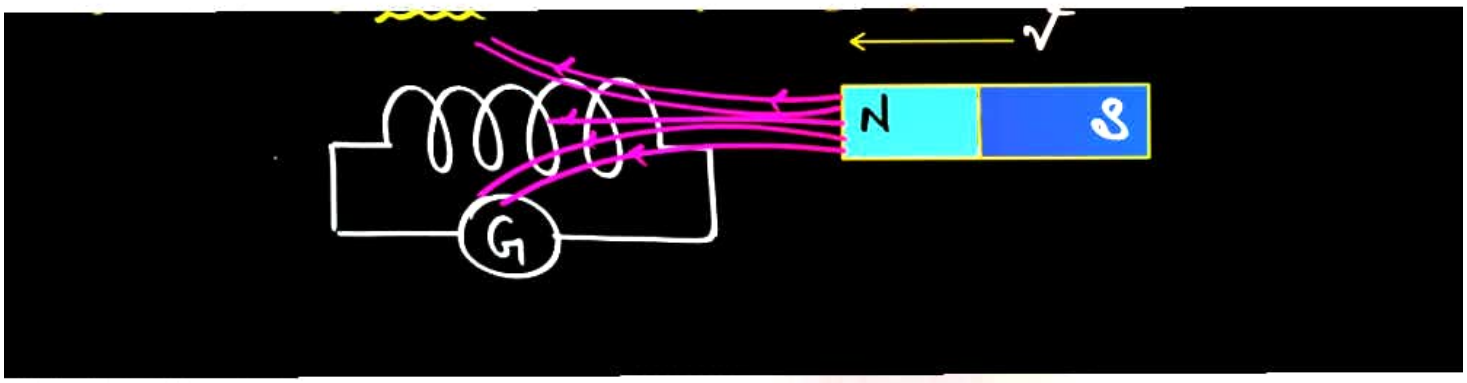
Faraday Law :-





:- EXPERIMENTS OF FARADAY AND HENARY :-

Experiment - 1 (current induced by a magnet)



(i) when north pole of a bar magnet is pushed towards the coil, the galvanometer shows a sudden deflection, which indicates that current is induced in the coil

(ii) There is no deflection when the bar magnet is held stationary anywhere.

(iii) when the magnet is moved away from the coil, the galvanometer shows, deflection in the opposite direction, which indicates reversal in the direction of induced curre

(iv) when south pole of bar magnet is moved towards on away, from the coil the galvanometer deflections are opposite to thos observed with the north pole for similar movement

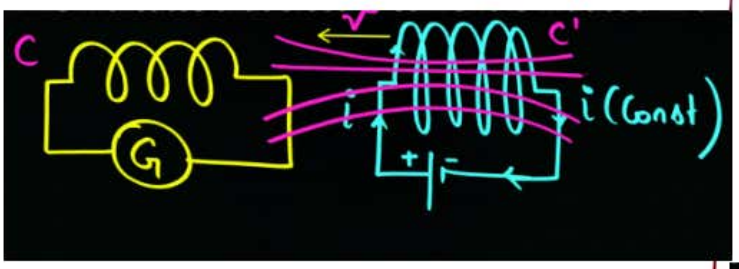
(v) The galvanometer deflection (and hence induced current) is found to be larger when magnet is pushed towards or pulled away from the coil faster.

(vi) when the bar magnet is held stationary and coil C is moved towards or away from the magnet, the same effects are observed.

It shows relative motion between the coil and the magnet is responsible for induction of electric current in the coil.

Experiment - 2
(Current induced by current)

(i) when coil C' is moved towards the coil C, the galvanometer shows a sudden deflection. This indicates that electric current is induced in the coil C.

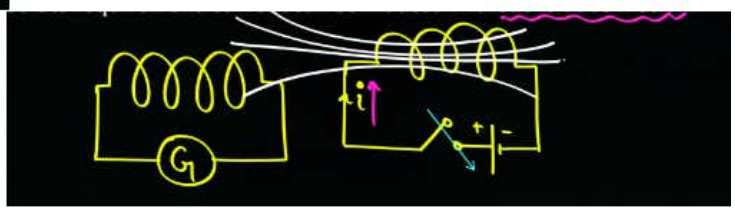


(ii) When the coil C' is moved away from the coil C, the galvanometer shows a deflection in the opposite direction. This indicates that direction of current induced in coil C is reversed.

(iii) There is no deflection when there is no relative motion between the two coils.

(iv) The galvanometer deflection (and hence induced current) is found to be larger when coils are moved faster towards /away from each other.

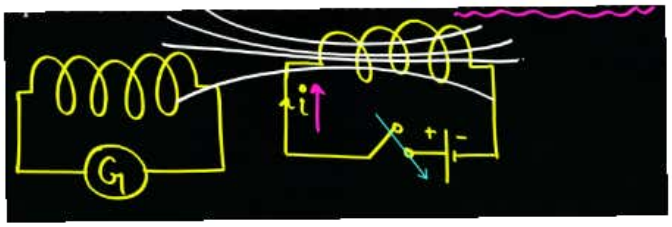
* Here the coil C' carrying current behaves as a bar magnet



Experiment – 3

(Current induced by Changing Current)

From the experiment 3 Faraday showed that relative motion is not an absolute requirement. Current can be induced without relative motion.

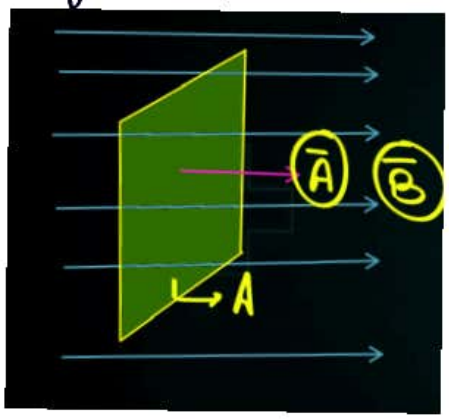


(i) when k key is pressed, galvanometer in coil C shows a momentary deflection, indicating that current is induced in coil C.

(ii) when the key is pressed continuously, there is no deflection in the galvanometer.

(iii) when the key is released the galvanometer shows again a momentary deflection but in the opposite direction.

(iv) The galvanometer deflection increases, when an iron rod is inserted into the coils along their axis, and the key is pressed/released.



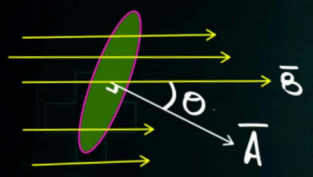
↳ induce current
↳ induce Emf
↳ change in flux
↳ Φ_B = No. of field passing per Unit Area

* $\Phi = \vec{B} \cdot \vec{A}$
$\Rightarrow |B||A| \cos\theta$ → B (M·F Const)

* $\Phi_B = \int B \cdot dA$
↳ (M·F variable)

$\phi_B = |B||A|\cos\theta$

θ is the angle b/w $\bar{B}$ & $\bar{A}$.



ϕ_B (Magnetic flux) is a Scalar Quantity

S.I Unit of Weber

Dimension of Magnetic flux? (Imp)

$F = iBl$

$B = \frac{f}{il}$

$B = \left[\frac{MLT^{-2}}{AL}\right] \Rightarrow \frac{MT^{-2}}{A}$

$\phi = MT^{-2}A^{-1} \times L^2$

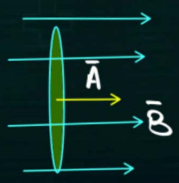
$\phi = [ML^2T^{-2}A^{-1}]$

$\phi = BA\cos\theta$

$\theta = 90°$, $\phi_B = 0$ (Min).

$\theta = 0°$, $\phi_B = |B||A|$ (Max)





ϕ → change

↳ Current

Ways to change flux?

$\phi_B \propto B$ (Magnetic field)

↳ motion, i ↑

$\phi_B \propto$ Area (Changing Area).

$\phi_B \propto (\theta)$ (Angle)



No. of field line (change)

Change in flux.

Induced Emf.

Induced Current.



Magnetic Flux (ϕ)

The magnetic flux (ϕ) through any surface held in a magnetic field $\underline{B}$ is measured by the total number of magnetic lines of force crossing the surface normally.

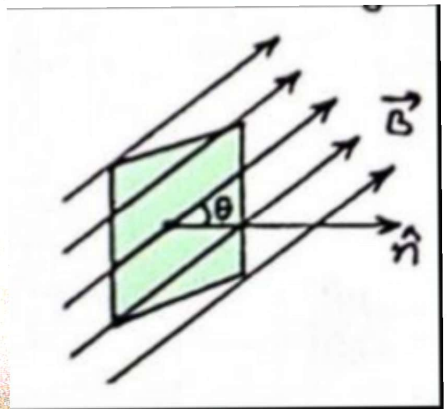
$$\boxed{\phi = \vec{B} \cdot \vec{A} = BA\cos\theta}$$

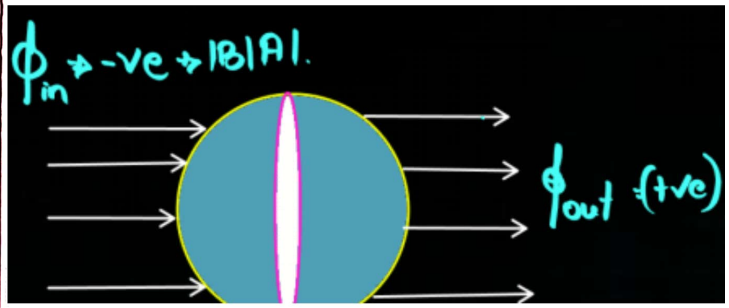
If the coil has N turns then total amount of magnetic flux linked with the coil is

$$\phi = N(\vec{B} \cdot \vec{A}) = NBA\cos\theta$$

* when $\theta = 90°$ then $\phi = 0$
  when $\theta = 0°$ then $\phi = BA$ (maximum value)

SI Unit of magnetic flux is weber (wb)





1 Weber (wb)–

One weber is the amount of magnetic flux over an area of 1 metre^2 held normal to a uniform magnetic field of 1 Tesla

$$\boxed{1 \text{ weber} = 1 \text{ tesla} \times 1 m^2}$$

magnetic flux (ϕ) is a scalar quantity

When a body is present in a uniform or non-uniform magnetic field, outward flux is taken to be positive, while inward flux is taken as negative.

Faraday's law of electromagnetic induction

$\Delta\phi$ = change in flux

$\mathcal{E}_{ind} = \frac{-d(\Delta\phi)}{dt}$ for N turns $\mathcal{E}_{ind} = \frac{-Nd(\Delta\phi)}{dt}$

Negative gives the dirn of Ind current (Lenz law)

*→ $\mathcal{E}_{ind} = \frac{-d(\Delta\phi)}{dt}$, $|\mathcal{E}_{ind}| = \frac{d\phi}{dt}$

$\Delta\phi$ = change → $(\phi_2 - \phi_1)$

$\mathcal{E}_{ind} = -\frac{\Delta\phi}{\Delta t} = -\frac{(\phi_2 - \phi_1)}{t_2 - t_1}$, $|\mathcal{E}_{ind}| = \frac{\Delta\phi}{\Delta t}$

for N turns $|\mathcal{E}_{ind}| = N\left(\frac{\Delta\phi}{\Delta t}\right)$

for N turns $|\mathcal{E}_{ind}| = \frac{N\times}{R}\left(\frac{\Delta\phi}{\Delta t}\right)$

*→ $I_{ind} = \frac{\mathcal{E}_{ind}}{R} = \frac{\Delta\phi}{R\Delta t} = \frac{\phi_2 - \phi_1}{R(t_2 - t_1)}$

First law :-

linked with whenever the amount of magnetic flux in the circuit changes, on e.m.f is induced as the circuit. The induced e.m.f exists so long as the change in magnetic flux continues

*→ As there is a change in flux there is induced emf. which produce induce current.

$\mathcal{E}_{ind} \propto -\frac{d\phi}{dt}$

*→ Charge

$q = \Delta t \cdot \rightarrow I(\Delta t)$

$q = \frac{\Delta\phi}{R \times \Delta t} \times \Delta t$

$q = \frac{\Delta\phi}{R} = \frac{\phi_2 - \phi_1}{R}$.

Second law :-

In a circuit is directly magnitude of e.m.f induced of change of magnetic flux linked with the circuit

induced emf $e \propto \frac{(\phi_2 - \phi_1)}{t}$, $e \propto \frac{\Delta\phi}{t}$

or $e = k\frac{(\phi_2 - \phi_1)}{t}$

Here $k = 1$ in all system of Units so

$$e = \frac{\phi_2 - \phi_1}{t}$$

OR

$$e = -\frac{d\phi}{dt}$$

Negative sign is taken because induced e.m.f. always apposes any change in magnetic flux associated with the circuit.

In case of a closely wound coil of N turns,

$$e = -N\frac{d\phi}{dt}$$

So by increasing the number of turns N in the coil we can increase the induced e.m.f.

If the total resistance of the closed loop (coil) is R then induced current $I = \frac{e}{R} \Rightarrow I = -\frac{N}{R}\frac{d\phi}{dt}$

The charge induced in the circuit in time Δt is given as

$$\because I = \frac{\Delta Q}{\Delta t} \Rightarrow I = -\frac{N}{R}\frac{\Delta\phi}{\Delta t} = \frac{\Delta Q}{\Delta t}$$

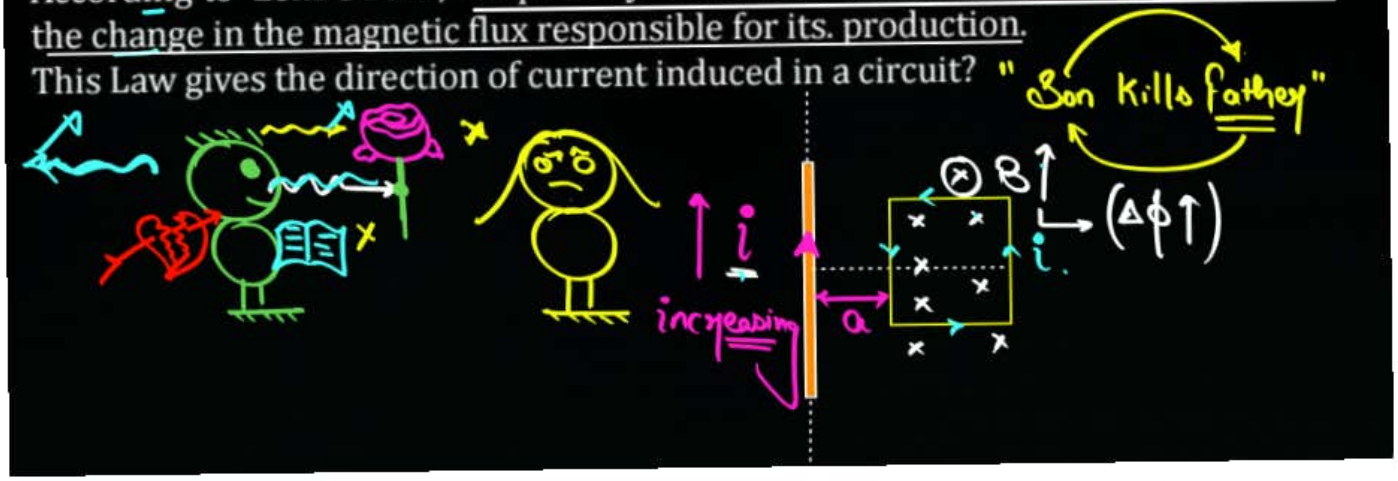
$$\Rightarrow \Delta Q = -\frac{N}{R}\Delta\phi$$

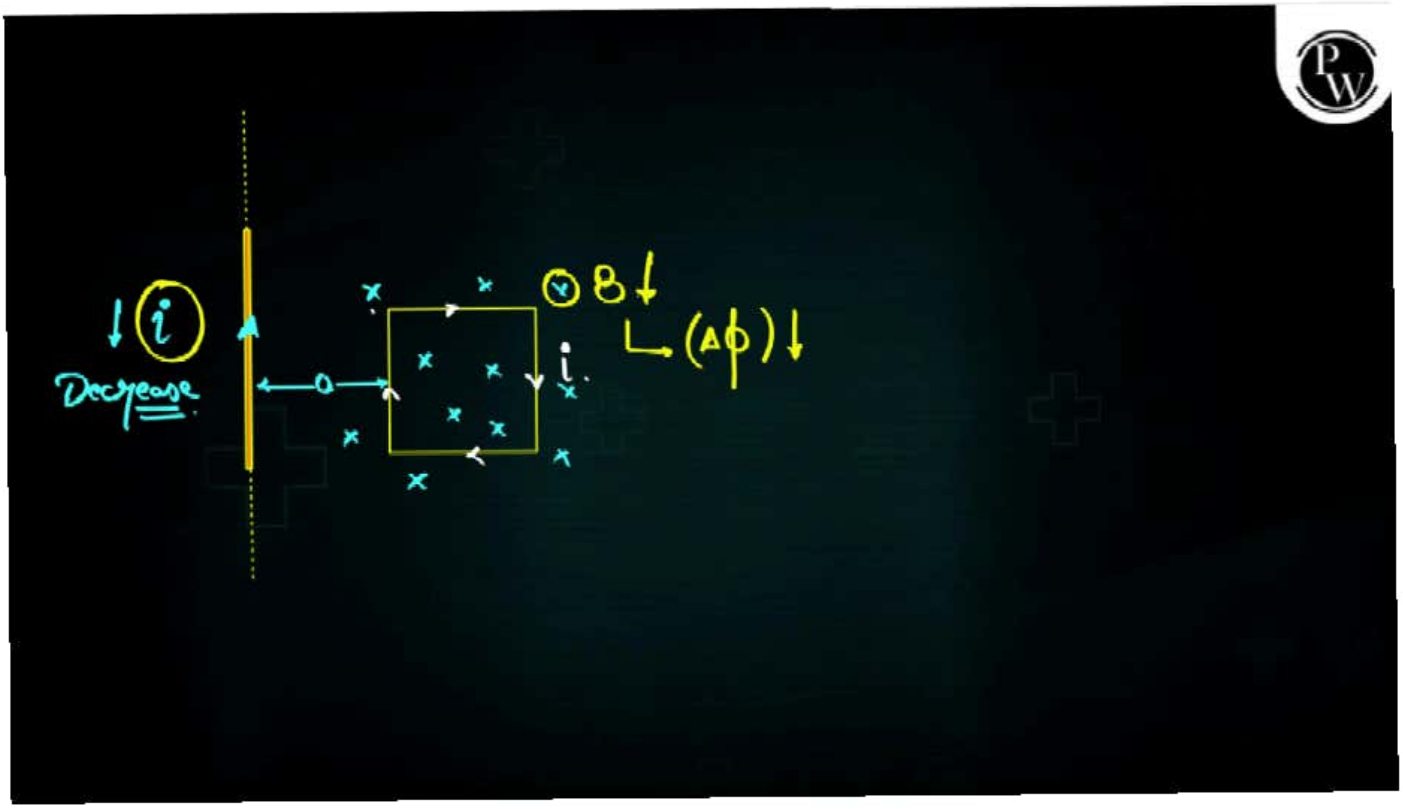
- Induced emf does not depends on nature of the coil and its resistance
- The induced charge depends on the net change in the magnetic flux and not on the time interval Δt of the flux change
- Induced current and induced charge depends on resistance of the coil (or circuit)

Lenz's law :-

According to 'Lenz's Law', the polarity of the induced em.f. is such that it opposes the change in the magnetic flux responsible for its. production.

This Law gives the direction of current induced in a circuit?





Explanation :-

When north pole of a bar magnet is being pushed towards the coil, the amount of magnetic flux linked with the coil increases.

Current is induced in the coil in such a direction that it opposes the increase in flux. This is possible only, when the current induced in the coil is in anticlockwise direction with respect to an observer on the side of the bar magnet

Similarly, when north pole of the bar magnet is moved away from the coil the magnetic flux linked with the coil decreases. The oppose this decrease in magnetic flux, current induced in the coil is in clockwise direction so that its face towards the bar magnet becomes south pole.

This would result in an attractive force, which opposes the motion of the magnet and the corresponding decrease in magnetic flux

Example

Example :-

Find the value of

1. $|\Delta\phi|$
2. $|\mathcal{E}_{ind}|$
3. $|i_{ind}|$
4. $q = ?$

Internal Resistance "r".



$\vec{B} = B_0 t^2 \hat{k}$

$\Delta\phi = |B| \cdot |A| \cos\theta$

$\theta = 0^\circ$

$\Delta\phi = B_0 t^2 \cdot \pi R^2$

$\Delta\phi = B_0 t^2 \pi R^2$

$|\mathcal{E}_{ind}| = \frac{d(B_0 t^2 \pi R^2)}{dt}$ → nx^{n-1}, $2x^{2-1}$.

$\Rightarrow B_0 \pi R^2 \frac{d(t^2)}{dt}$

$|\mathcal{E}_{ind}| = B_0 \pi R^2 \times 2t$

③ $i_{ind} = \frac{|\mathcal{E}_{ind}|}{r} = \frac{2B_0 \pi R^2 t}{r}$, $q = It$ ↳ I = Const.

④ $i = f(\text{time})$.

$$\int dq = \int I \cdot dt = \int \frac{2B_0 \pi R^2}{r} \cdot t \cdot dt$$

$$\Rightarrow \frac{2B_0 \pi R^2}{r} \int t \cdot dt \Rightarrow \frac{2 \times B_0 \pi R^2 \times t^2}{2r} = \frac{B_0 \pi R^2 t^2}{r}$$

$\Rightarrow \frac{x^{n+1}}{n+1}$

Example

Example :-

Find the value of

1. $|\Delta\phi|$ $(R = R_0 t)$
2. $|\mathcal{E}_{ind}|$
3. $|i_{ind}|$
4. $q = ?$

Internal Resistance "η".



1. $\Delta\phi = |B||A|\cos\theta$

$\Delta\phi = B_0 \times \pi (R)^2 \cos 0^\circ$

$\Delta\phi = B_0 \times \pi (R_0 t)^2$

$\Delta\phi = B_0 \pi R_0^2 t^2$

2. $|\mathcal{E}_{ind}| = \frac{d(\Delta\phi)}{dt} = B_0 \pi R_0^2 \frac{d(t^2)}{dt}$

$\Rightarrow 2 B_0 \pi R_0^2 t.$

3. $i_{ind} = \frac{|\mathcal{E}_{ind}|}{\eta} = \frac{2 B_0 \pi R_0^2 t}{\eta}$

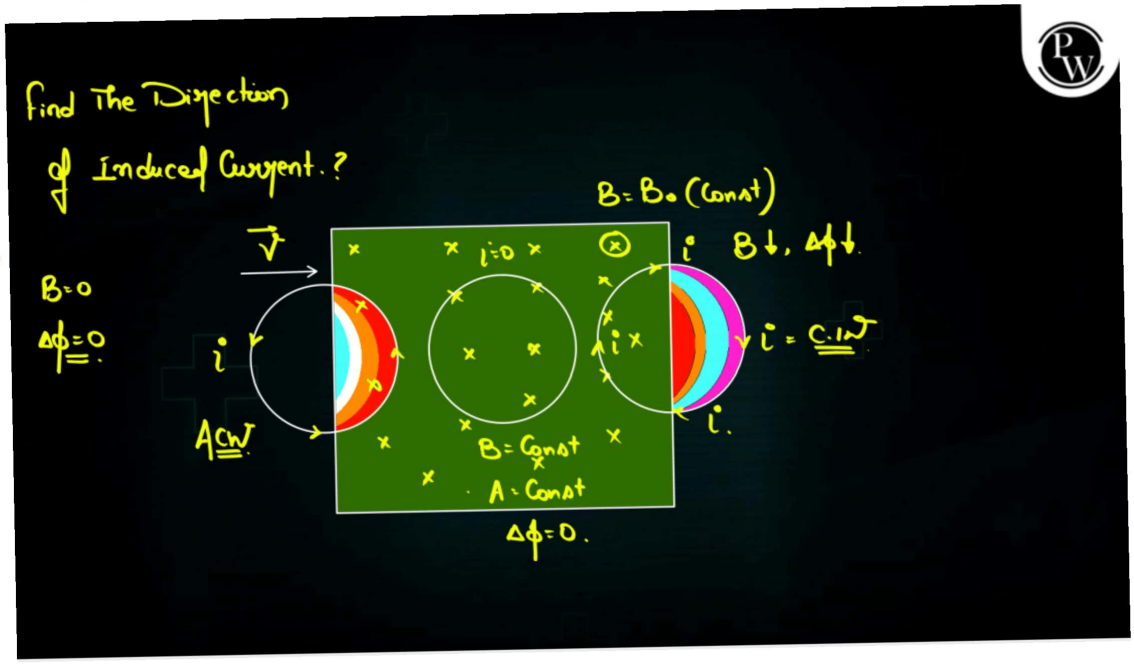
4. $i = f(\text{time})$

$$\int dq = \int i\,dt = \int \frac{2 B_0 \pi R_0^2}{\eta} t\,dt$$

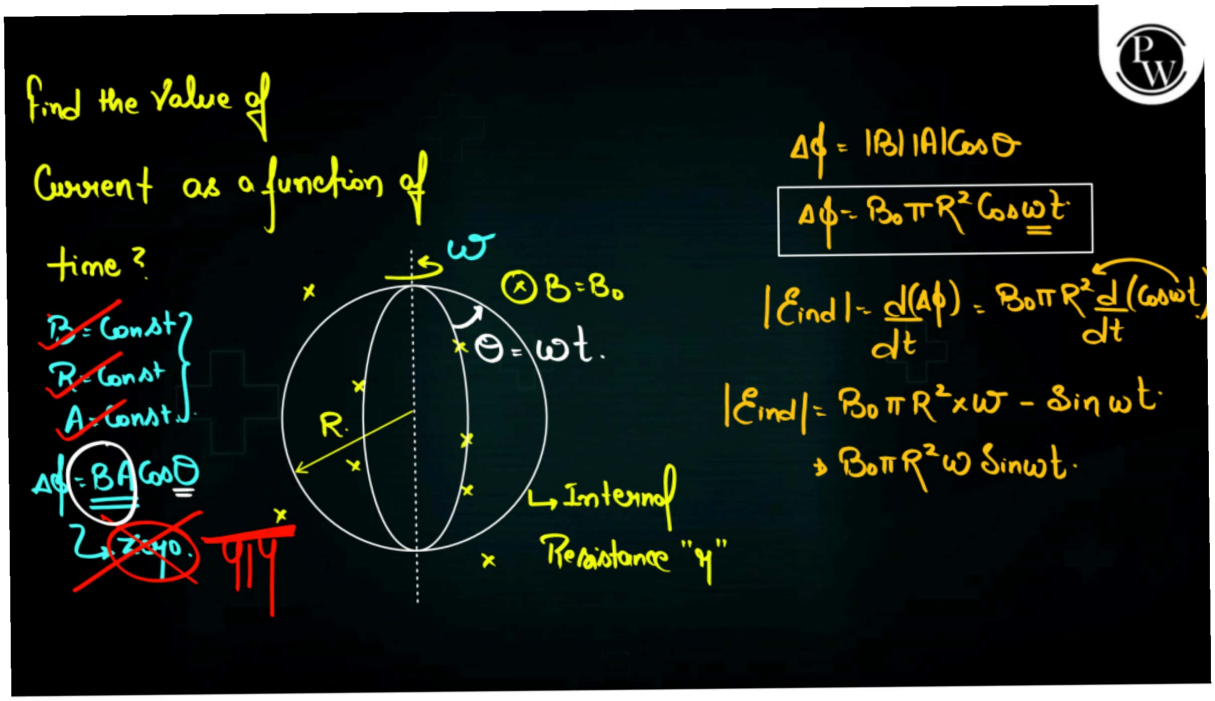
$$q = \frac{2 B_0 \pi R_0^2}{\eta} \int_0^t t\,dt \rightarrow \frac{2 B_0 \pi R_0^2}{\eta} \times \frac{t^2}{2}$$

$$q = \frac{B_0 \pi R_0^2 t}{\eta}$$

Find the Direction of Induced Current :-



Find the value of Current as a function of time?

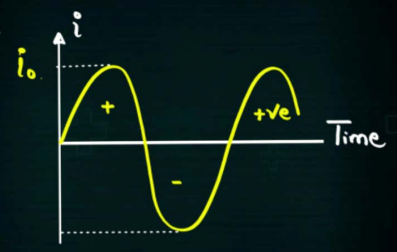


$i = \frac{|\mathcal{E}_{ind}|}{\mathcal{R}} = \frac{B_0 \pi R^2 \omega \sin\omega t}{\mathcal{R}}$

$i = \left(\frac{B_0 \pi R^2 \omega}{\mathcal{R}}\right) \sin\omega t$

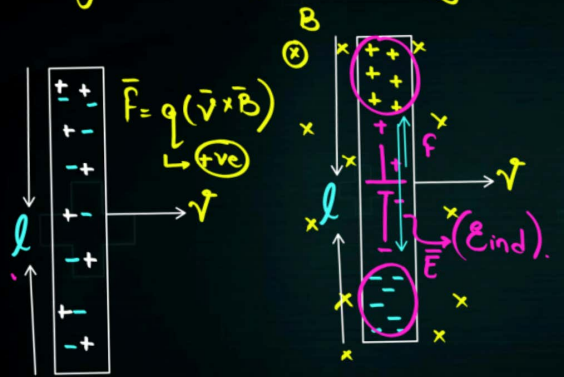
$\boxed{i = i_0 \sin\omega t}$

$i = f(\sin\omega t)$



Induced Emf due straight Rod (moving)

Induced Emf due Straight Rod (moving).



$f_E = f_B$

$q\bar{E} = q(\bar{v} \times \bar{B})$

$\bar{E} = (\bar{v} \times \bar{B})$ → velocity.

$\Delta V = \bar{E} \cdot \bar{\ell}$ → length of Rod

$\mathcal{E}_{ind} = \bar{E} \times \ell$

$\mathcal{E}_{ind} = (\bar{v} \times \bar{B})$

ℓ

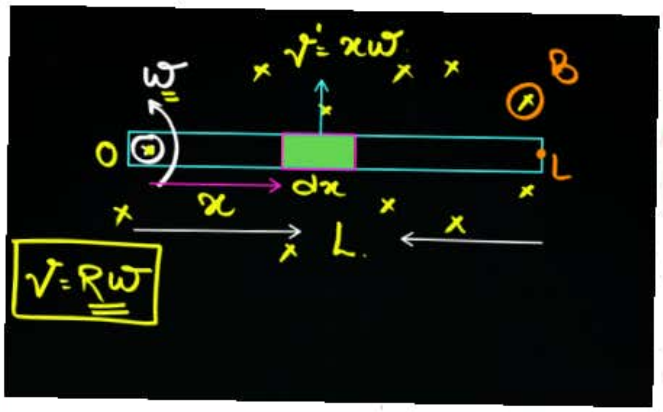
$\boxed{\mathcal{E}_{ind} = \bar{\ell} \cdot (\bar{v} \times \bar{B})}$

$\mathcal{E}_{ind} = vB\ell \sin\theta$

$\bar{v} \perp \bar{B}$.

$\boxed{\mathcal{E}_{ind} = vB\ell}$

Induced EMF due to rotation of a conducting rod in uniform magnetic field :-



$$\mathcal{E}_{ind} = \bar{l} \cdot (\bar{v} \times \bar{B})$$

$$\int dv = \int dx \cdot x\omega B$$

$$V = \mathcal{E}_{ind} = B\omega \int_0^L x \cdot dx$$

$$\mathcal{E}_{ind} = B\omega \left(\frac{x^2}{2}\right)_0^L$$

$$\boxed{\mathcal{E}_{ind} = \frac{B\omega L^2}{2}}$$

Energy consideration in motinal E.M.F.

when a conductor of length L, is moved with a velocity v in a perpendicular magnetic field B, then the motional e.m.f produced in the conductor is

$$e = BLV$$

Let r be the resistance of movable arm PQ of the rectangular conductor. Here we assume that the resistance of remaining arm QR, RS and SP is negligible compared to r.

Current induced in the loop

$$I = \frac{e}{r} = \frac{BLV}{r}$$

The magnitude of force on the conductor PQ moving in the magnetic field is

$$F = BIL = B\left(\frac{BLV}{r}\right)L = \frac{B^2L^2V}{r}$$

The direction of this force is opposite to the velocity of the conductor.

Power required to push the conductor is

$$P_m = F \times V = \frac{B^2L^2V}{r} \times V = \frac{B^2L^2V^2}{r}$$

As the conductor is pushed mechanically, the machanical energy dissipated per second is given by

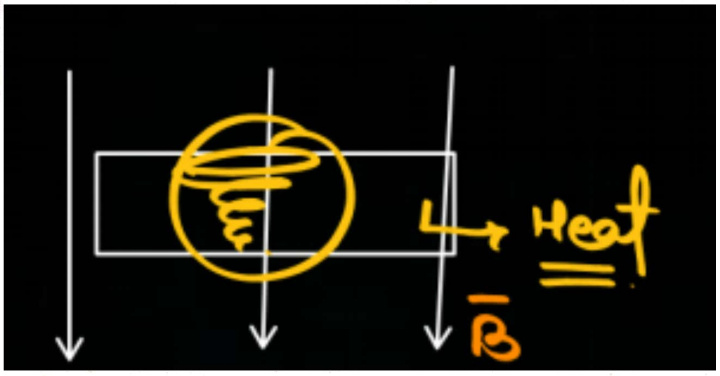
$$P_E = I^2 r = \left(\frac{BLV}{r}\right)^2 r = \frac{B^2L^2V^2}{r}$$

which is the same as the power required to push the conductor.

Hence, mechanical energy required to move the conductor PQ is converted into electrical energy first (i.e the induced e.m.f) and then to thermal energy

$$\text{Heat energy produced/sec} = \frac{B^2L^2V^2}{r}$$

EDDY CURRENTS



(The eddy currents are basically the currents induced in the body of a conductor due to change in magnetic flux linked with the conductor.)

For example when we move a metal plate out of a magnetic field, then the relative motion of the field and the conductor induces a current in the conductor. induced current whirl about within the plate as it they were caught in an eddy (or whirlpool) of water. This is called <u>eddy current</u>.

Eddy current (i) = $\frac{\text{Induced e.m.f (E)}}{\text{resistance (R)}}$

But we know that

$e = -\frac{d\phi}{dt}$ so

$$i = -\frac{d\phi/dt}{R}$$

The direction of eddy current is given by Lenz's law or Fleming's right hand rule

Self Induction

"Solenoid"



It is the property of a coil by virtue of which the coil opposes any change in the strength of current flowing through it by inducing an e.m.f in itself.

When current (I) is increasing, the self induced e.m.f. (e) appears across the coil in a direction such that is opposes the increase. Therefore it would be in a direction opposite to I.

$\phi_B \propto I$

$\phi_B = LI$

$N\phi_B = LI$

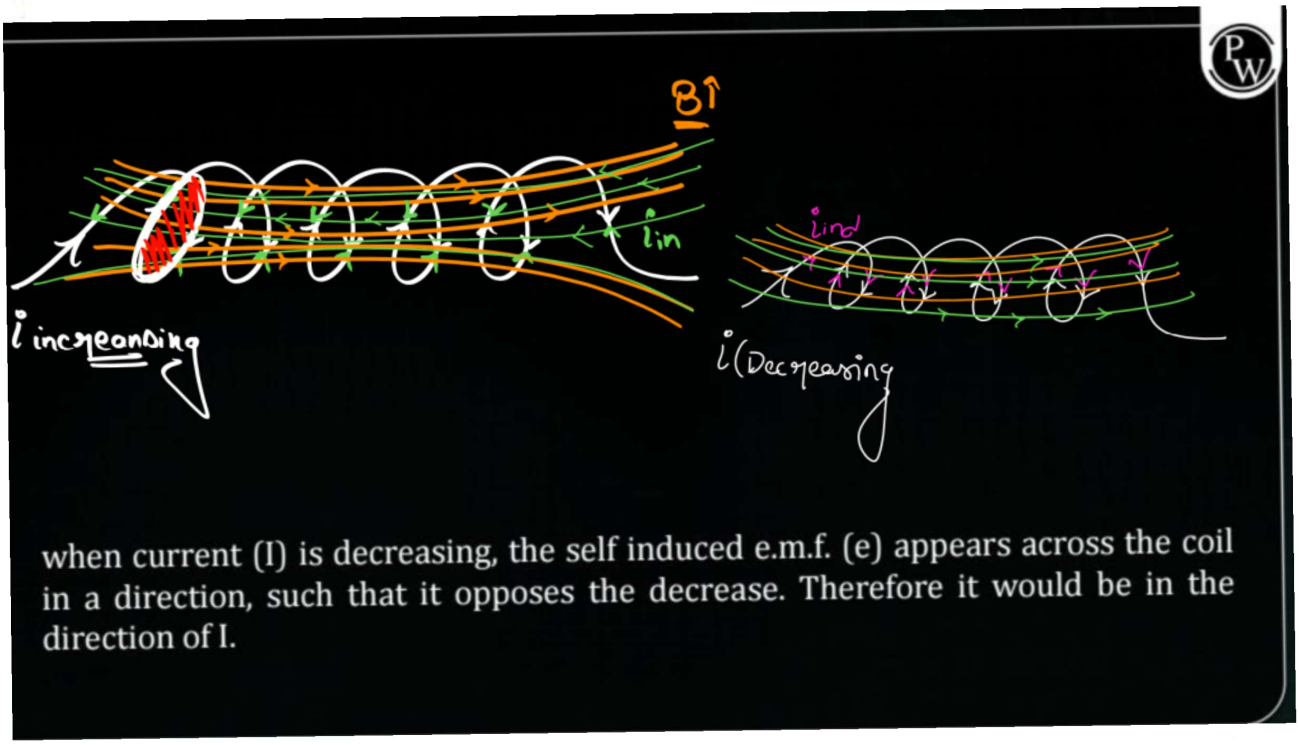
$\phi_B = \frac{LI}{N}$

$\mathcal{E}_{ind} = -\frac{d\phi_B}{dt}$

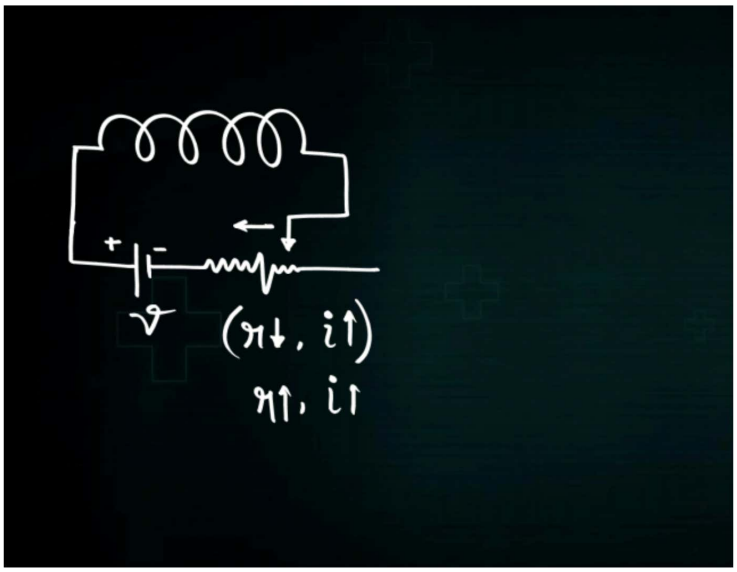
$\mathcal{E}_{ind} = -\frac{Ldi}{dt}$

$\mathcal{E}_{ind} = -\frac{d(Li)}{dt}$

L = co-efficient of self inductance



when current (I) is decreasing, the self induced e.m.f. (e) appears across the coil in a direction, such that it opposes the decrease. Therefore it would be in the direction of I.



$B_{solenoid} = \mu_0 n i$

$n = \frac{N}{L}$

$\phi = B \cdot A$

$\phi_B = LI = (\mu_0 n i)\pi R^2$

$L\not{I} = \mu_0 \frac{N}{\ell} \times \not{I} \times \pi R^2$

$$\boxed{L = \frac{\mu_0 N \pi R^2}{\ell}}$$

$L \propto N$

$L \propto R$

$L \propto \frac{1}{\ell}$

Self Induction :-

Let the current I flowing through a coil at any time and ϕ is the amount of magnetic flux linked with all the turns of the coil at that time. Then it is found that

$$\phi \propto I \quad \text{or} \quad \phi = LI$$

Here L is a constant of proportionality and is called self inductance of coefficient of self induction of the coil

- The value of L depends upon the number of turns, area of section and nature of material of the core on which the coil is wound.

If $I = 1$, $\phi = L \times 1$ or $L = \phi$

Self inductance of a coil is numerically equal to the amount of magnetic flux linked with the coil when unit current flows through the coil.

The e.m.f. induced in the coil is given by.

$$e = -\frac{d\phi}{dt} = -\frac{d}{dt}(LI)$$

$$e = -L \times \frac{di}{dt}$$

The SI unit of L is henry

$$L = \frac{-e}{di/dt}$$

$$\therefore 1 \text{ henry} = \frac{1 \text{ volt}}{1 \text{ ampere/sec}}$$

The self inductance of a coil is said to be 1 henry when a current change at the rate of 1 ampere/sec through the coil induces an e.m.f. of 1 volt in the coil

Self inductance of a solenoid

Consider a long air solenoid having number of turns per unit length n. and current flowing in it is I.

where $n = \frac{N}{I}$

magnetic field within the Solenoid $B = \mu_0 nI$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m

If A is cross-sectional area of solenoid, then flux linked with solenoid of length I is

$$\phi = NBA\cos 0^\circ$$

$$\phi = N \times \mu_0 nl = N \times \mu_0 \frac{NIA}{l}$$

$$\phi = \frac{\mu_0 N^2 AI}{l}$$

Self inductance of air solenoid

$$L = \frac{\phi}{I}$$

$$L = \frac{\mu_0 N^2 A}{l}$$

If solenoid contains a core of ferromagnetic substance of relative permeability μ_r, then self inductance

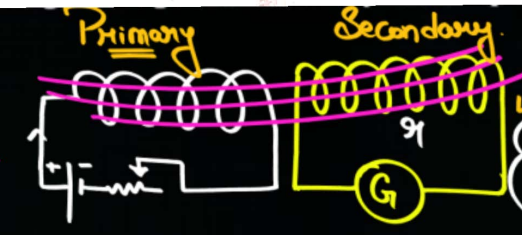
$$L = \frac{\mu_r \mu_0 N^2 A}{l}$$

Mutual Inductance

It is the property of two coils by virtual of which each, opposes any change in the strength of current flowing through the other by developing an induced e.m.f.

$\phi_s \propto i_p$

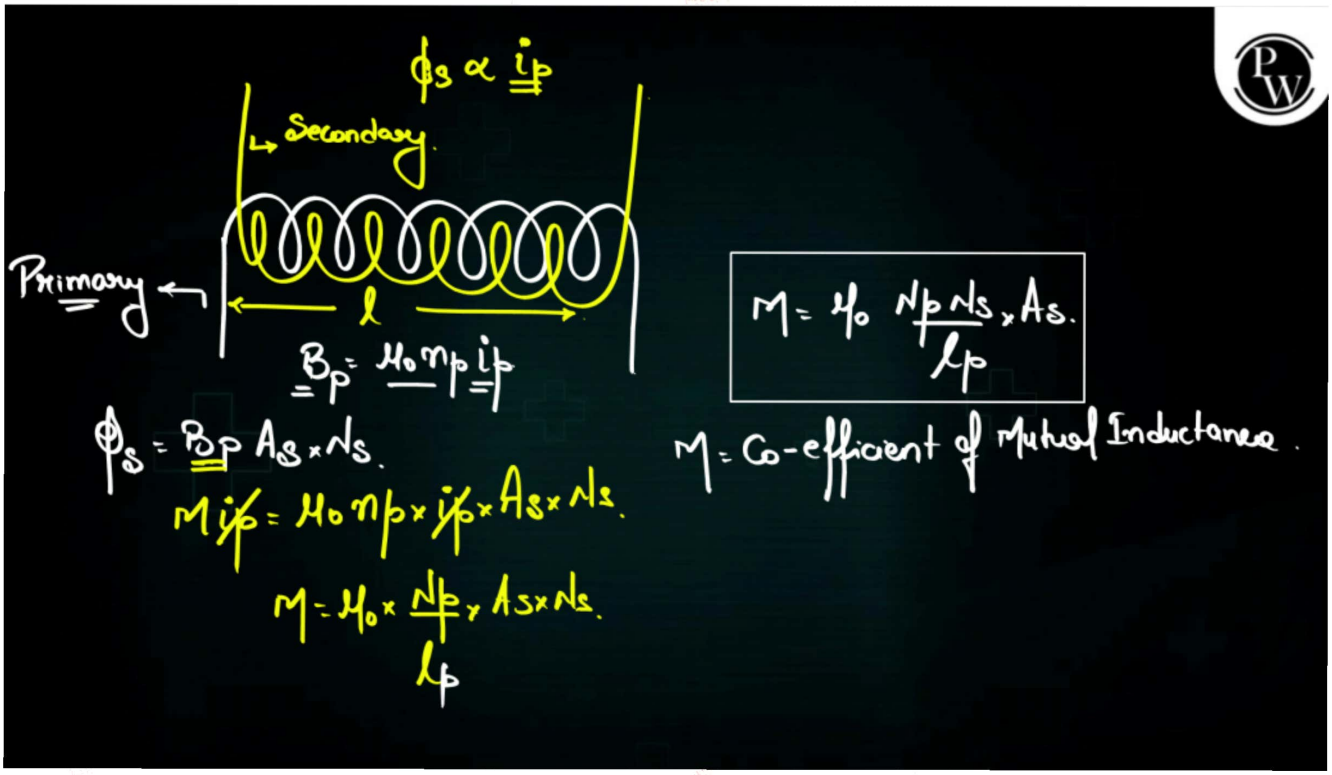
$\phi_s = M i_p$



$(\mathcal{E}_{ind})_s = -\frac{d(\Delta\phi_s)}{dt}$

$(\mathcal{E}_{ind})_s = -M\frac{d(i_p)}{dt}$

$I_s = \frac{(\mathcal{E}_{ind})_s}{\mathcal{R}}$



$\phi_s = B_p A_s \times N_s$

$M i_p = \mu_0 n_p \times i_p \times A_s \times N_s$

$M = \mu_0 \times \frac{N_p}{l_p} \times A_s \times N_s$

$M = \mu_0 \frac{N_p N_s}{l_p} \times A_s$

M = Co-efficient of Mutual Inductance.

On pressing or releasing K, galvanometer shows a sudden temporary deflection.

This is due to mutual induction as detailed below-

(i) On pressing key k, current in P increase from, zero to maximum value it takes some time. During this time, current in P is increasing. Therefore magnetic flux linked with P is increasing

As S is closeby, magnetic flux associated with S also increases. An e.m.f. is induced in S. According to lenz's law, the induced current in S would oppose increase in current

in P by flowing in a direction opposite to the cell current in P

(ii) On releasing key K, current in P decreases from maximum to zero value. It takes some time. During this time current in P is decreasing. Therefore magnetic flux linked with P is decreasing. As S is held close by, magnetic flux associated with S also decreases. An e.m.f is induced in S

According to lenz's law induced current in S during break flows in the direction of the cell current in P so as to oppose the decrease in current in P.

Mutual Inductance:-

It is the property of two coils by virtue of which each, opposes any change in the strength of current flowing through the other by developing an induced e.m.f.

Let I is the strength of current in one coil and ϕ is the total amount of magnetic flux linked with all the turns of the neighbouring coil. Then it is found that

$$\phi_2 \propto I_1 \text{ or } \phi_2 = MI_1$$

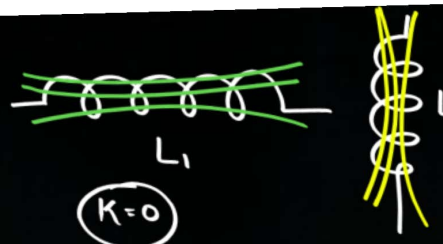
where M is a constant of proportionality and is called co-efficient of mutual induction or mutual inductance of the two coils.

Coefficient of Coupling

$M = K\sqrt{L_1 L_2}$

$K = \dfrac{M}{\sqrt{L_1 L_2}}$

$0 < K < 1$



$K = 0$

$K = 1$

↳ Ferro Magnetic Sub.

$\int x^n \to \dfrac{x^{n+1}}{n+1} \to \dfrac{x^{1+1}}{1+1} \to \dfrac{x^2}{2}$

Energy Stored



L → $\mathcal{E}_{ind} = L\dfrac{di}{dt}$

$P = Vi$

$\text{Power} = \dfrac{\text{Energy}}{\text{Time}} = \dfrac{dE}{dt}$



$P\,dt = dE$

$V\,i\,dt = dE$

$Li\dfrac{di}{dt} \times dt = dE$

$\int dE = \int L\,i\,di$

$E = L \times \dfrac{i^2}{2}$

$\boxed{E = \frac{1}{2}Li^2}$

Energy Density → $\dfrac{\text{Energy}}{\text{Volume}}$

$B = \mu_0 n i$

$L = \mu_0 n^2 (\pi R^2 \times L)$ ↳ Vol.

$L = \mu_0 n^2 \times \text{Vol}$

$E = \frac{1}{2} \times \mu_0 n^2 \times \text{Vol} \times i^2$

$\dfrac{E}{\text{vol}} = \dfrac{\mu_0 n^2 i^2}{2} \times \dfrac{\mu_0}{\mu_0}$

$\dfrac{E}{\text{vol}} = \dfrac{\mu_0^2 n^2 i^2}{2\mu_0}$

$\boxed{\text{Energy Density} = \dfrac{B^2}{2\mu_0}}$